

Two-Camera IR Mocap Hover

Frames

$[\cdot]^W$: world (ArUco marker centre, $+x$, $+y$ marker edges, $+z$ up)

$[\cdot]^B$: body ($+x$ nose, $+y$ portside, $+z$ up) (1)

$[\cdot]^C$: camera ($+x$ right, $+y$ down, $+z$ optical axis)

$[a]_{\times}$: skew matrix, $[a]_{\times} b = a \times b$

$\text{Exp}(\phi)$: rotation of angle $\|\phi\|$ about $\phi/\|\phi\|$ (Rodrigues)

$\pi(\cdot)$: perspective divide, $\pi\left(\begin{pmatrix} a \\ b \\ c \end{pmatrix}\right) = \begin{pmatrix} a/c \\ b/c \end{pmatrix}$

Camera pose

Problem

One flat marker fixes the world frame; each camera solves its own pose against it.

$$X_j = (\pm L/2, \pm L/2, 0)^\top, \quad j = 1 \dots 4 \quad (2)$$

$$s \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = K[R|t] \begin{pmatrix} X_j \\ 1 \end{pmatrix} \quad (3)$$

X_j : marker corners in W [m] L : marker edge [m]

x_j : measured corner pixels [px] K : intrinsics

$R \in \text{SO}(3)$, t : world $\rightarrow$ camera pose

Solution

$$\hat{R}, \hat{t} = \arg \min_{R, t} \sum_{j=1}^4 \|\pi(K(RX_j + t)) - x_j\|^2 \quad (4)$$

Closed form for a planar square (IPPE), which returns both mirror poses; keep the smaller residual, report the ratio as an ambiguity check.

$$C = -R^\top t \quad (5)$$

C : camera centre in W [m]

Drone pose

Triangulation

Per LED, from $x \times PX = 0$, two rows per camera.

$$AX = 0, \quad A = \begin{pmatrix} u_0 p_0^{3\top} - p_0^{1\top} \\ v_0 p_0^{3\top} - p_0^{2\top} \\ u_1 p_1^{3\top} - p_1^{1\top} \\ v_1 p_1^{3\top} - p_1^{2\top} \end{pmatrix} \quad (6)$$

$P_i = K[R|t]$: projection matrix, camera i $p_i^{k\top}$: its row k

x_0, x_1 : undistorted pixels of one LED [px]

X : right singular vector of A at $\sigma_{\min}$, dehomogenised

Correspondence

Blobs arrive unlabelled; $3! = 6$ pairings, take the smallest mean reprojection error.

$$\sigma^* = \arg \min_{\sigma \in S_3} \frac{1}{3} \sum_{k=1}^3 \sum_{i=0}^1 \left\| \pi(P_i X_{k,\sigma}) - x_{i,k} \right\|^2 \quad (7)$$

Rigid fit (Kabsch)

$$H = \sum_{k=1}^3 (b_k - \bar{b})(w_k - \bar{w})^\top = U \Sigma V^\top \quad (8)$$

$$R = V \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(VU^\top) \end{pmatrix} U^\top, \quad p = \bar{w} - R\bar{b} \quad (9)$$

$$\text{rmsd} = \sqrt{\frac{1}{3} \sum_{k=1}^3 \|Rb_k + p - w_k\|^2} \quad (10)$$

b_k : LED positions in B [m] w_k : triangulated LEDs in W [m]
 $\det(VU^\top)$ term: forces a rotation, not a reflection
rmsd: fit residual [m], also the quality gate
labelling of w_k against b_k : chosen by the same minimisation

LQR control law

Problem

One axis, error $e = z - z_{\text{ref}}$. Betaflight holds attitude; the commanded lean arrives through a first-order lag.

$$x = \begin{pmatrix} \int e \\ e \\ \dot{e} \\ a \end{pmatrix}, \quad u = a_{\text{cmd}}, \quad \dot{x} = Ax + Bu \quad (11)$$

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1/\tau \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1/\tau \end{pmatrix} \quad (12)$$

$$J = \sum_{k=0}^{\infty} x_k^\top Q x_k + u_k^\top R u_k \quad (13)$$

a : acceleration actually delivered, chasing u [m/s²]
 τ : attitude (or throttle) lag + transport delay [s]
 $Q = \text{diag}(1/x_{i,\text{max}}^2)$, $R = \rho^2/a_{\text{max}}^2$: Bryson weights
 ρ : effort, trades aggression against smoothness

Solution

Zero-order hold at the loop rate, then the discrete Riccati equation.

$$\begin{pmatrix} A_d & B_d \\ 0 & I \end{pmatrix} = \exp\left(\begin{pmatrix} A & B \\ 0 & 0 \end{pmatrix} \Delta t\right) \quad (14)$$

$$P = A_d^\top P A_d - A_d^\top P B_d (R + B_d^\top P B_d)^{-1} B_d^\top P A_d + Q \quad (15)$$

$$K = (R + B_d^\top P B_d)^{-1} B_d^\top P A_d, \quad u = -Kx \quad (16)$$

P : solved by iterating the recursion to convergence (no SciPy)

Command mapping

World demand $\rightarrow$ body by measured yaw ψ , then to sticks.

$$a_{\text{fwd}} = a_x \cos \psi + a_y \sin \psi, \quad a_{\text{left}} = -a_x \sin \psi + a_y \cos \psi \quad (17)$$

$$\theta = \arctan(a_{\text{fwd}}/g), \quad \varphi = \arctan(-a_{\text{left}}/g), \quad \text{thr} = \text{ff} (1 + a_z/g) / \cos \theta \quad (18)$$

θ, φ : pitch, roll commands [rad], clamped to θ_{max}

ff: hover feed-forward, learned in flight from the standing vertical integral

yaw: left to the flight controller

EKF

State

Nominal (p, v, R, ω) kept as is; the filter propagates the error state, so rotation needs no global parameterisation.

$$\delta x = \begin{pmatrix} \delta p \\ \delta v \\ \delta \theta \\ \delta \omega \end{pmatrix} \in \mathbb{R}^{12}, \quad R_{\text{true}} = R \text{Exp}(\delta \theta) \quad (19)$$

p, v : position, velocity in W [m], [m/s]

R, ω : body $\rightarrow$ world rotation, body rate [rad/s]

error on the right: keeps H simple, no Euler singularity

Prediction

Constant velocity, constant body rate; acceleration is process noise.

$$p^- \leftarrow p + v \Delta t, \quad R^- \leftarrow R \text{Exp}(\omega \Delta t), \quad P^- \leftarrow F P F^\top + Q \quad (20)$$

$$F = \begin{pmatrix} I & I \Delta t & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & \text{Exp}(\omega \Delta t)^\top & I \Delta t \\ 0 & 0 & 0 & I \end{pmatrix} \quad (21)$$

Measurement model

Three LEDs, $z \in \mathbb{R}^9$.

$$h_k = R b_k + p, \quad H_k = [I \ 0 \ -R[b_k]_\times \ 0] \quad (22)$$

z : triangulated LED positions [m], labelled against the **prediction**

$R_m = \sigma_m^2 I_9$: triangulation noise

Update

Gate first, then Joseph form.

$$y = z - h, \quad S = H P^- H^\top + R_m, \quad y^\top S^{-1} y \leq \chi_{\text{gate}}^2 \quad (23)$$

$$K = P^- H^\top S^{-1}, \quad \delta x = K y \quad (24)$$

$$P^+ \leftarrow (I - K H) P^- (I - K H)^\top + K R_m K^\top \quad (25)$$

$$p^+ = p^- + \delta p, \quad v^+ = v^- + \delta v, \quad R^+ = R^- \text{Exp}(\delta \theta), \quad \omega^+ = \omega^- + \delta \omega \quad (26)$$

y : innovation S : innovation covariance K : Kalman gain

gate rejected, or no measurement: predict only, invalid past t_{coast}

R^+ : re-orthonormalised by SVD